Saiyam Bajpai

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Education

Madhav Institute of Technology and Science, Gwalior — Gwalior, Madhya Pradesh
B.Tech, Computer Science and Design, Minor in Design — Aug 2025 – May 2029

Indian Institute of Technology, Madras — Chennai, Tamil Nadu
BS, Data Science and Applications (Remote) — Aug 2026 – May 2030

Skills

Languages: Python, C/C++, SQL, JavaScript, HTML/CSS
ML & Data Tools: Scikit-Learn, SHAP, YOLOv8, Ultralytics, OpenCV, MediaPipe, NumPy, Pandas, Matplotlib
Frameworks: React, Node.js, Flask, FastAPI
Developer Tools: Git, Docker, Google Cloud Platform, Google Colab, MongoDB, Ollama
Concepts: Computer Vision, Object Detection, Explainable AI (XAI), Risk Modeling, Data Preprocessing

Projects

FinCA XAI – Credit Analyser | *Python, Scikit-Learn, SHAP, Risk Modeling* | Aug 2026 – Present
Github

- Built a credit scoring platform integrating machine learning and Explainable AI (SHAP) to eliminate algorithmic bias in credit risk assessments.
- Combined predictive risk modeling algorithms with institutional review dashboards and financial eligibility calculators.

Pothole Detection Model | *Python, YOLOv8, OpenCV, Numpy, Colab (T4)* | Dec 2025 – Mar 2026
Github

- Trained a YOLOv8 object-detection model using the Ultralytics framework on Google Colab to identify potholes in road-surface images.
- Preprocessed and augmented a custom dataset of 20,000 images with OpenCV and NumPy to handle varied lighting and road conditions.
- Achieved 75.9% precision, 68.4% recall, and 0.75 mAP@50 on validation batches through hyperparameter tuning.

GestureFlow | *Python, MediaPipe, OpenCV, NumPy, Git* | Feb 2026 – June 2026
Github

- Built a gesture-based control system using Python, MediaPipe, and OpenCV to enable hands-free scrolling, volume, and brightness adjustments.
- Engineered 6 documented iterations to incrementally expand gesture capabilities, scaling from a CLI-driven terminal system (v1–v5) to a web-enabled application in v6.
- Deployed the system for local execution on Windows without external hardware dependencies.

Experience & Achievements

National Summer Hackathon SkillStation — 2026
Machine Learning Lead (Team of 4)

- Trained the core machine learning pipeline for QueueLess Bharat to provide an automated queue-management solution.